Apply Two-Variable Relationships to Solve Problems

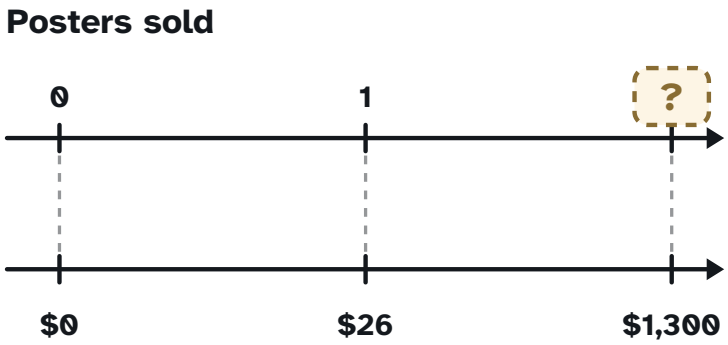
Start from a goal and work backward with an equation to find the input.

Name _____ **Date** _____

Tonight's spark

A band earns $26 for each poster it sells. Its goal is $1,300. How many posters must the band sell to reach the goal?

Guess first: more than 30 posters, or fewer?



Our class model: work backward from the goal

Try it

Work it out on paper. Write one answer on each line.

1 **In e = 26n, the goal is e = 130. Which equation do you solve?**

A. 26n = 130 **B.** 26 + n = 130 **C.** n = 130 × 26 **D.** e = 26

My answer _____

2 **Solve 26n = 130.**

A. 104 **B.** 5 **C.** 156 **D.** 3,380

My answer _____

3 **What undoes multiplying by 9?**

A. Adding 9 **B.** Subtracting 9 **C.** Dividing by 9 **D.** Multiplying by 9

My answer _____

4 **A savings plan is s = 12w and the goal is $96. Which equation finds the weeks w?**

A. 12 + w = 96 **B.** w = 96 + 12 **C.** 96w = 12 **D.** 12w = 96

My answer _____

5 **Solve 12w = 96.**

A. 84 **B.** 8 **C.** 108 **D.** 1,152

My answer _____

6 **Check w = 8 by putting it back into s = 12w. What is s?**

A. 20 **B.** 4 **C.** 96 **D.** 108

My answer _____

Talk about it

What is your goal, and where did you put it in the equation? Which operation undid the multiplication?

Home mission

Pretend you save $4 each week toward a $48 goal. Can your student write 4w = 48 and undo the multiplying? Then put the answer back: do that many weeks of $4 really make $48?

If we get stuck, ask

- What is the goal, and where does it go in the equation?
- What was done to the variable? What undoes that?
- Put your answer back in. Does it reach the goal?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/9-4-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it